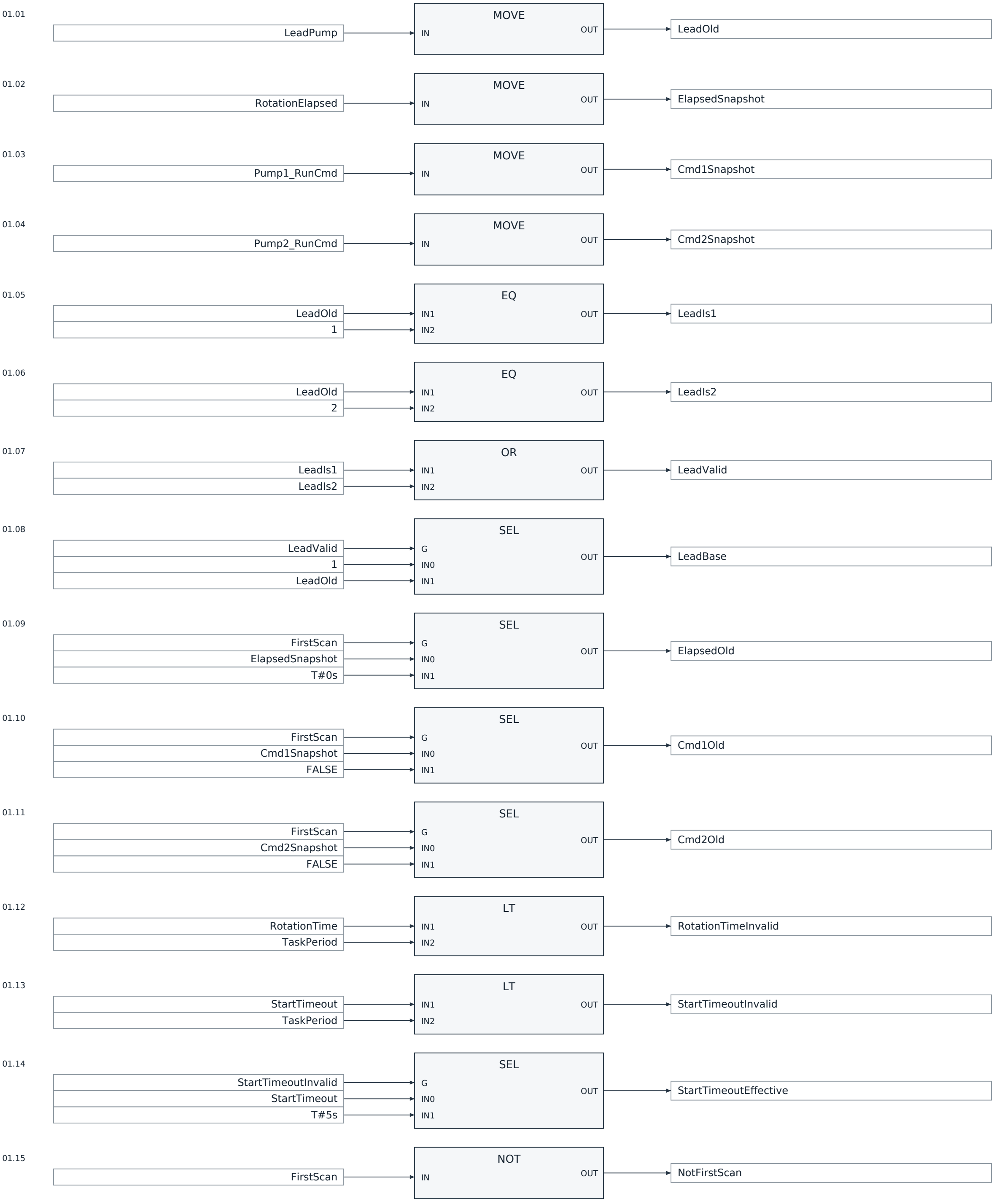
01 | Snapshot / LeadPump normalization / parameter checks

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

Execute top to bottom. Snapshot state BEFORE later networks write it. FirstScan is supplied by the PLC platform.

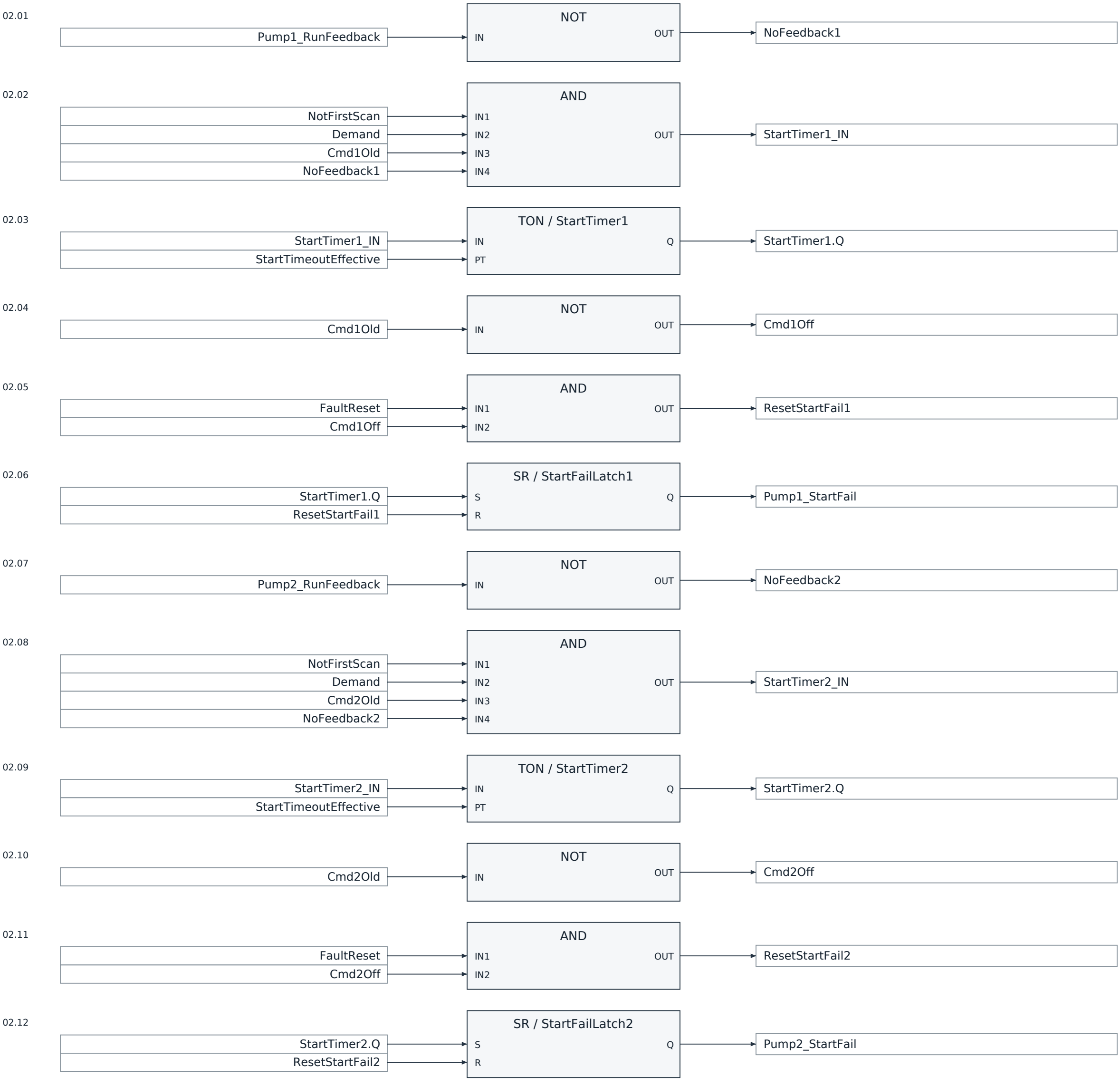


02 | Start / feedback supervision and latched StartFail

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

SR is SET-dominant. Reset is accepted only when the previous RUN command is OFF. TON also detects sustained feedback loss.



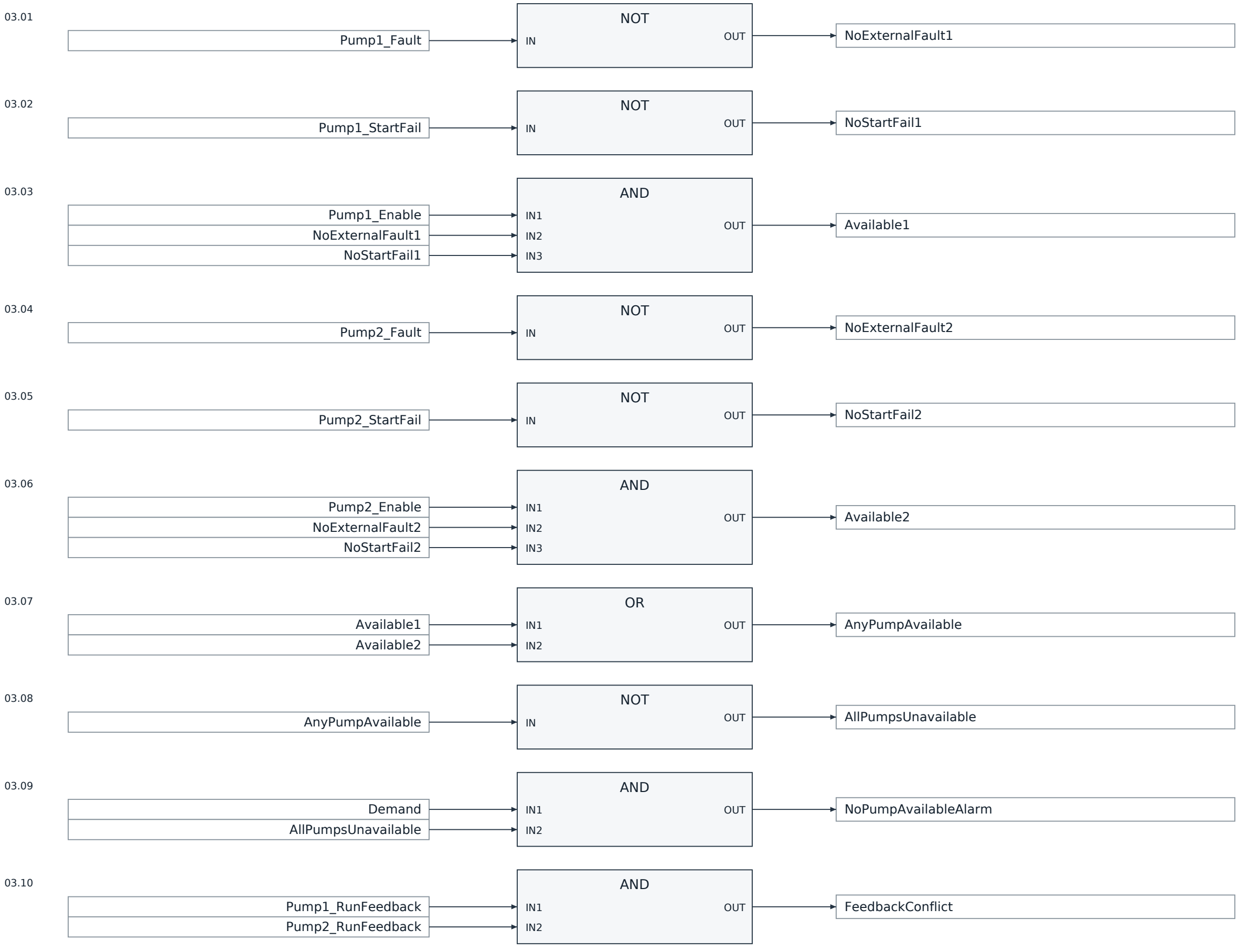
Network 2/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.

03 | Pump availability / unavailable alarms / feedback conflict

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

AllPumpsUnavailable is independent of Demand. NoPumpAvailableAlarm requires Demand.



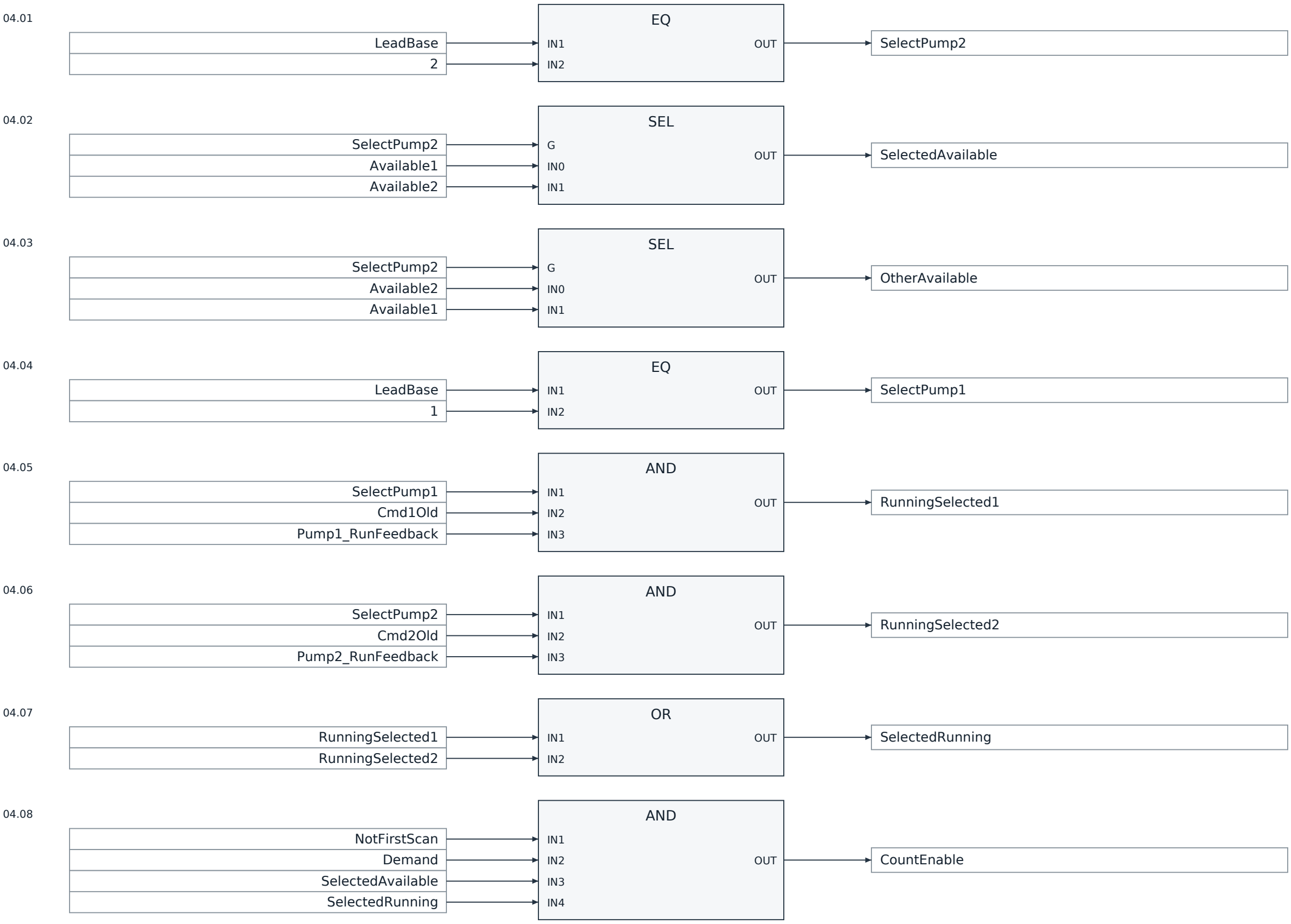
Network 3/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.

04 | Selected / standby pump and confirmed-running condition

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

Time accumulation requires previous RUN command AND the matching RunFeedback. RunCmd alone is insufficient.



05 | ForceRotation rising edge

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

One pulse per rising edge. A request rejected because the standby pump is unavailable is NOT queued.

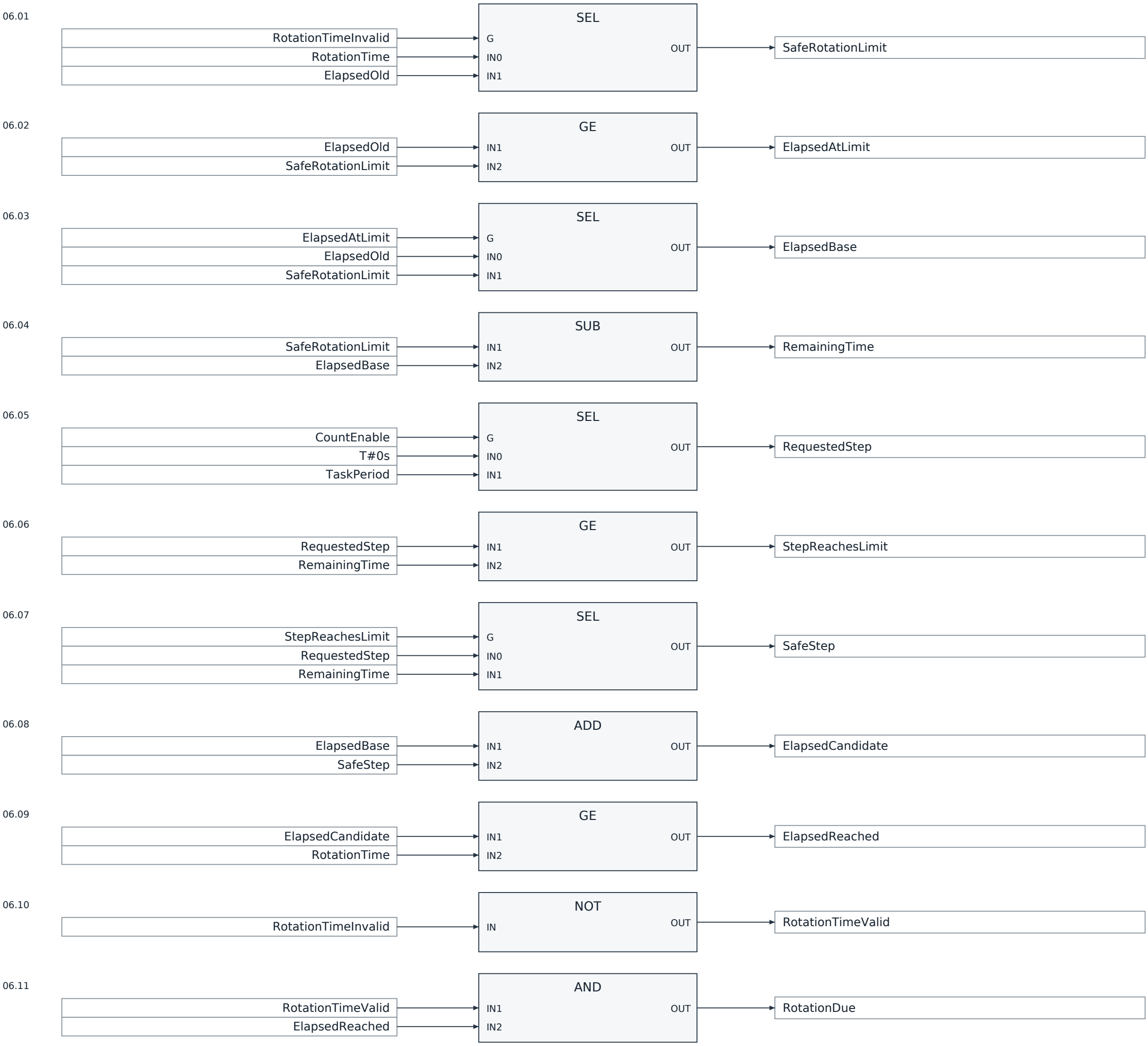
Network 5/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.

06 | Accumulated rotation timer / TIME arithmetic / RotationDue

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

100 ms cyclic task; TaskPeriod = T#100ms. Demand gaps HOLD time. Saturation prevents overflow. Invalid RotationTime disables timed rotation.



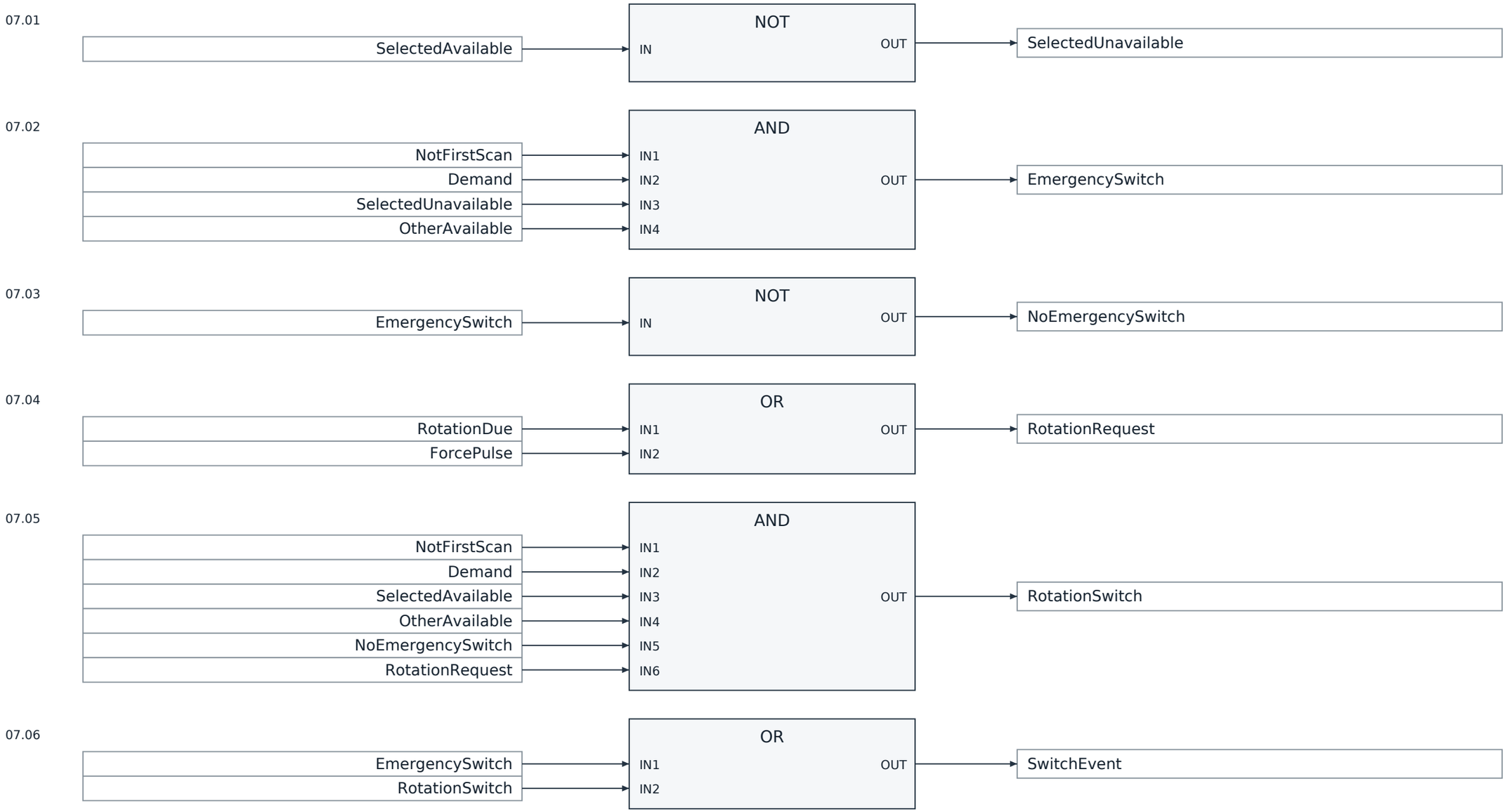
Network 6/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.

07 | Emergency switching / timed and forced rotation arbitration

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

Emergency switching is independent of the rotation timer. Only one switch event is evaluated from LeadBase per scan.



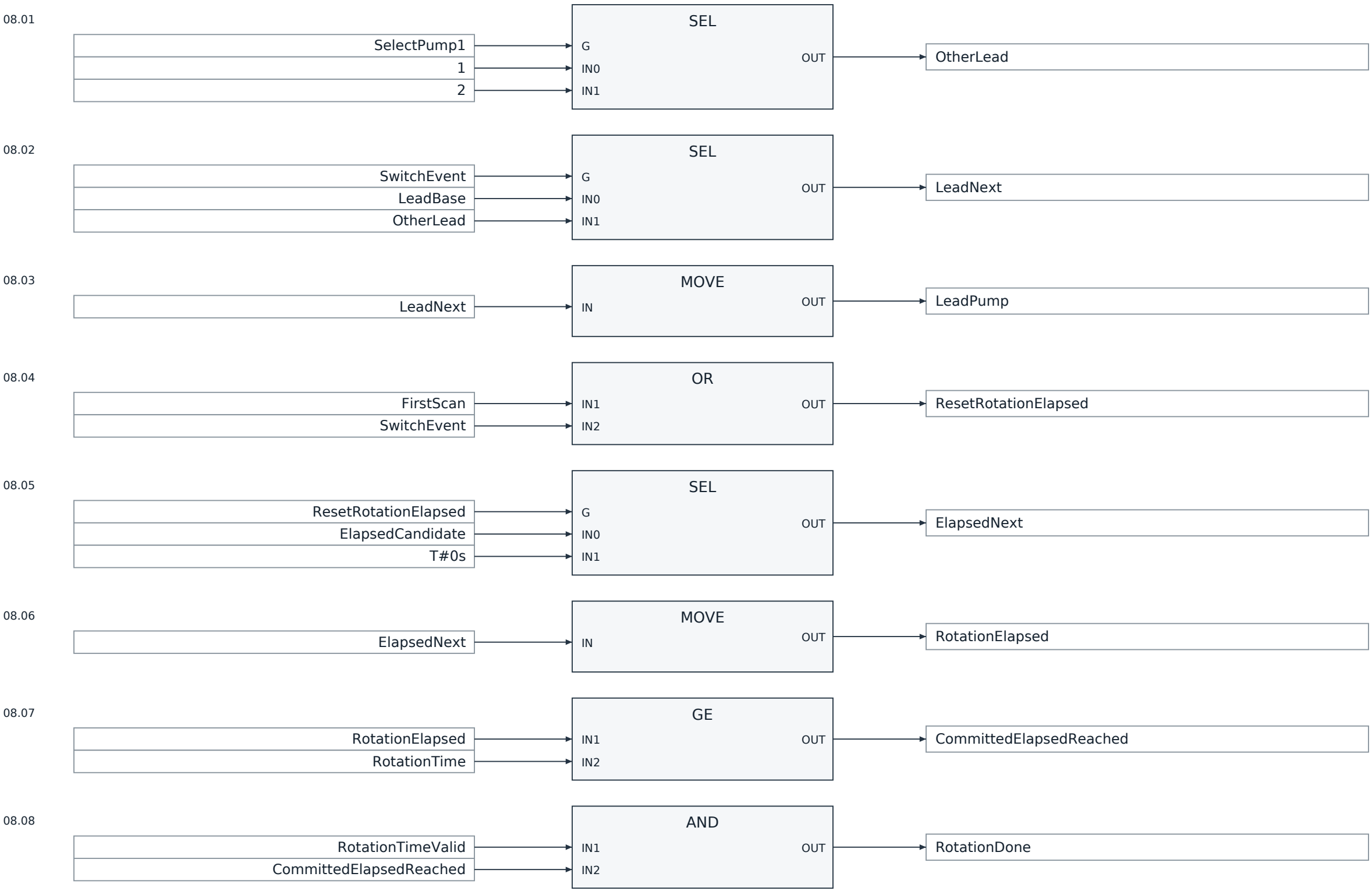
Network 7/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.

08 | Commit LeadPump / RotationElapsed reset or hold / RotationDone

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

LeadPump is RETAIN. RotationElapsed is NOT RETAIN: reset on FirstScan or switching; otherwise commit the accumulated value.



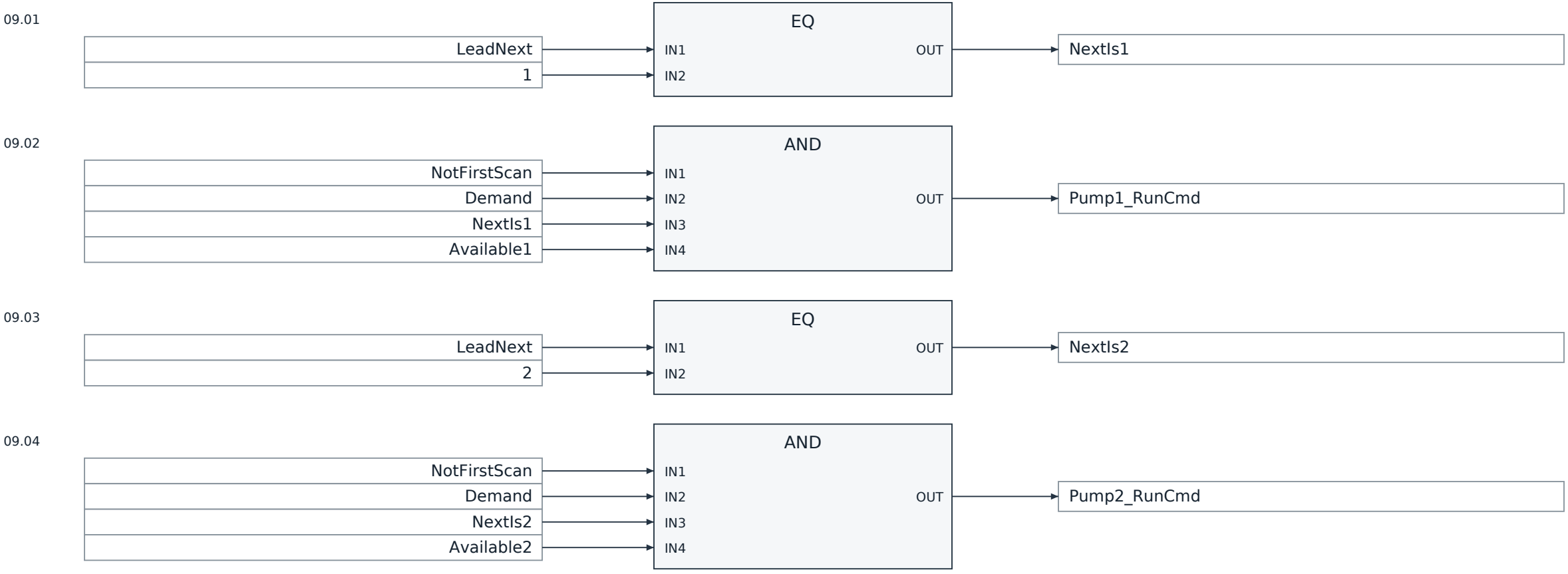
Network 8/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.

09 | RUN outputs - exclusive single assignment

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN

RotationTime: adjustable TIME | TaskPeriod: T#100ms | SEL: FALSE selects IN0; TRUE selects IN1

The ONLY writers of Pump1_RunCmd and Pump2_RunCmd. EQ(LeadNext,1) and EQ(LeadNext,2) cannot both be TRUE.



Network 9/9 | Named boxes are signal connectors; identical names are connected. Arrows show data flow. IEC review diagram; not compiled for TM172.